Agentic AI-Based Intelligent Customer Support and Order Management System for Online Clothing Stores

Primary Advisor: Rabranea Bqa

M. Ebrahim Khan
261945465

Adeel Liaqat
261032149

Hashim Aleem
271047457

ABSTRACT

The rapid growth of online shopping has created a need for more efficient and intelligent customer support systems, particularly in the clothing retail sector. Traditional support methods often struggle to manage large volumes of customer queries, order processing, and personalized recommendations. This project proposes the development of an ***Intelligent Customer Support and Order Management System Using Agentic AI for Online Clothing Stores****. The system will feature a multilingual AI assistant capable of handling customer queries in both Urdu and English through text and voice interactions. It will allow users to place and track orders conversationally while receiving fashion and clothing recommendations based on preferences and occasions. A budget optimization module will help users find products within their price range. Additionally, sentiment analysis will detect customer complaints, and a smart admin dashboard will provide insights into orders, customer interactions, and product trends.*

1. INTRODUCTION

The rapid growth of e-commerce has significantly transformed the way people purchase clothing, making online shopping more convenient, accessible, and time-efficient. However, despite these advancements, managing customer interactions, handling orders, and delivering personalized shopping experiences remain major challenges for many online clothing stores. Traditional customer support systems largely depend on human agents, which often results in delayed responses, inconsistent service quality, and inefficiencies in order processing. As the number of online shoppers continues to rise, businesses face increasing pressure to handle large volumes of customer queries while maintaining high levels of customer satisfaction and engagement. In this context, Artificial Intelligence (AI) offers innovative solutions by enabling automation, intelligent decision-making, and real-time customer interaction. This project proposes the development of an Intelligent Customer Support and Order Management System using Agentic AI specifically for online clothing stores. The system will feature multilingual customer support in both Urdu and English, allowing users to communicate through text and voice interfaces. It will enable customers to place orders conversationally and track their orders in real time, improving convenience and usability. Additionally, the system will incorporate AI-based fashion recommendations tailored to user preferences, occasions, and trends, along with budget optimization to assist customers in making cost-effective choices. Furthermore, sentiment analysis will be used to automatically detect customer complaints and feedback, while a smart admin dashboard will provide valuable insights into customer behavior, order statistics, and product performance. Overall, the proposed system aims to enhance customer experience, improve operational efficiency, and support data-driven decision-making in the online clothing retail sector.

2. PROBLEM STATEMENT

Online clothing stores are rapidly growing due to the increasing popularity of e-commerce; however, they continue to face significant challenges in delivering efficient customer service and managing orders effectively. Traditional customer support systems rely heavily on human agents, which often leads to delayed responses, increased workload, and a higher chance of human errors in order processing. These limitations negatively impact customer satisfaction, especially when users expect quick and accurate assistance during their shopping experience. Furthermore, customers frequently struggle to find clothing items that match their personal preferences, occasions, and budget constraints, as many platforms lack intelligent recommendation systems tailored to the clothing domain.

3. LITERATURE REVIEW

The rapid development of machine learning and artificial intelligence (AI) technology has significantly changed how companies interact with consumers in the online market. In order to improve overall shopping experiences, improve product recommendations, and automate customer service, modern e-commerce platforms are depending more and more on intelligent technologies. For online organizations, technologies like sentiment analysis, recommendation systems, conversational AI, and analytics dashboards have become essential. These technologies can help companies in reducing operating costs, speed up response times, and providing clients personalized support. Because of this, researchers and tech companies are always looking for ways to combine various technologies into unified system that enhance customer experience and efficiency in operations.

There are already a number of platforms in the industry that provide chatbot-based customer engagement tools for businesses. **Zendesk, Intercom, and LiveChat,** for example: provide AI-assisted customer support solutions that let companies respond to queries from customers via automated chat platforms. Similar to this, conversational AI companies like **Omilia** offer voice-activated virtual assistants that allow businesses communicate with clients automatically via text and voice. Businesses can use messaging apps like WhatsApp to interact with clients via chat services like **Yalo and Wati.** While these systems offer efficient customer communication solutions, the majority of them focus on chatbot-based support and messaging services. They frequently lack integrated features like sentiment analysis, multilingual communication, intelligent product recommendations, and comprehensive administrative analytics within a single, unified system.

Conversational AI for customer service is one of the most widely researched topics in this field. Conversational agents, commonly referred to as chatbots, use machine learning (ML) and natural language processing (NLP) techniques to simulate human-like conversation. Conversational AI systems can greatly enhance customer service in e-commerce environments by automating answers to frequently asked questions about products, order status, and return policies, according to **Singh and Sharma [1].** According to their research, intelligent chatbots can retain effective and reliable customer communication while reducing response times and organizational workload. Similarly, conversational AI frameworks can revolutionize digital commerce by facilitating more organic and context-aware interactions between user and online platforms, according to **Hussain and Lemon [2].** According to their research, conversational systems can enhance user interaction and produce personalized customer experiences.

Another important aspect of modern conversational systems is multilingual communication, particularly in areas where many languages are commonly spoken. Accessibility for users who desire to communicate in their native languages is limited by the reality that the majority of commercial chatbot systems are mainly English-speaking. In order to overcome this difficulty, **Awais and Nawab [3]** suggested a method for improving Urdu text-to-speech and voice cloning technologies. Their research highlights the importance of language-specific AI models in making intelligent systems accessible to regional users. Similar to this, **Ahmed and Ali [4]** used the Rasa framework to create a multilingual Urdu chatbot, showing how natural language processing methods may be used to understand and deal with Urdu queries using transliteration and language modelling. According to their research, in areas where English is not the primary language of communication, multilingual conversational systems can greatly increase user accessibility and engagement.

The development of recommendation systems is a significant topic of study in e-commerce technology. In order to recommend products that are most relevant to customers, recommendation systems examine user preferences and behavioral data. In online retail platforms, these technologies are crucial for increasing product sales and improving customer experience. **Reddy, Kumar, and Singh [5]** created a smart fashion recommendation system that analyzes clothing photos and recommends visually similar fashion items using deep learning models like **CNN and ResNet-50**. Their research shows how computer vision methods might improve the shopping experience by helping customers in finding outfits that suit their personal preferences. In the same way, **Gupta and Malhotra [6]** created a fashion recommendation system that uses deep learning algorithms to produce personalized product recommendations based on browsing behavior and product requirements.According to their research, advanced recommendation systems in online shopping platforms may increase client engagement and purchase likelihood.

Sentiment analysis has emerged as an essential techni for understanding customer feedback and improving service quality, in addition to recommendation systems. Sentiment analysis determines if customer feedback are neutral, negative, or positive by analyzing textual comments using machine learning algorithms. In order to examine customer reviews on e-commerce platforms, **Wang and Li [7]** developed a sentiment analysis model based on the **BiLSTM** neural network. Their research shows that deep learning methods can improve sentiment classification accuracy and successfully extract contextual information from customer feedback. For sentiment analysis of e-commerce product reviews, **Chen, Wu, and Zhang [8]** suggested a hybrid **CNN-LSTM** model.Their approach improves the accuracy of identifying emotional patterns in textual data by combining long short-term memory networks for sequence analysis with convolutional neural networks for feature extraction.

The usage of analytics tools and administrative dashboards is another essential component of intelligent e-commerce platforms. Companies can manage interactions with customers, keep an eye on system performance, and analyze sales data in real time using these dashboards. Graphs, charts, and performance indicators are examples of visual analytics that help company managers in understanding operational trends and making defensible decisions. However, companies frequently have to rely on several technologies for analytics dashboards, sentiment analysis, recommendation systems, and customer support. This separation makes the system more complicated and makes it more difficult to manage and integrate data from many sources.

Even while AI-based customer support and e-commerce technologies have advanced significantly, many of the platforms that are now in use focus on certain features rather than offering a comprehensive solution. For example, some systems focus on recommendation systems or analytics tools, while others specialize in chatbot-based consumer communication. Additionally, many commercial systems still have limited multilingual support for regional languages like Urdu. Similar to this, a lot of fashion recommendation models may not adequately account for regional cultural preferences or customer's budget constraints because they are trained on datasets that mainly represent Western fashion styles.

Based on these observations, there is a clear need for a single platform that combines several AI technologies into one system. Thus, our project proposes the development of a Multilingual AI Customer Support and Order Management System with Fashion Recommendation and Sentiment Analysis. The goal of the proposed system is to improve business management and customer experience by combining multiple intelligent features into a single platform.

An AI-powered chatbot that can answer customer queries regarding orders, products, and general support issues will be part of the system. Customers will be able to interact with the system in their preferred language due to its capability for multilingual communication in both Urdu and English. Additionally, the platform will have an intelligent fashion recommendation system that makes recommendations for relevant clothing based on user likes and browsing activities. Customers will be able to receive product recommendations that meet their budget limitations and fashion preferences due to the budget-aware recommendations feature.

Sentiment analysis for customer feedback, which analyzes customer complaints and reviews to determine emotional sentiment, is another important component of the proposed method. Companies will be able to identify dissatisfied consumers and improve service quality with the help of this feature. Additionally, the platform will have an order management system that allow customers to place orders, track order status. Finally, an administrative dashboard with visual analytics will provide business management information about system performance, product trends, and customer interactions.

The proposed system aims to fix the problems of present systems and offer an integrated AI-driven e-commerce system by integrating multiple technologies into a single platform. This integrated system will improve business decision-making, increase customer engagement, and develop a more intelligent and effective shopping environment.

Table of comparisons of applications that already exist and our proposed application (Agentic AI-Based Intelligent Customer Support and Order Management System for Online Clothing Stores)

Feature	Zendesk	Intercom	LiveChat	Omilia	Yalo	Wati	Existing E-Commerce Platforms	Proposed System
AI-Based Customer Support	✓	✓	✓	✓	✓	✓	**Limited**	✓
Multilingual Support (Urdu + English)	×	×	×	×	×	×	**Partial**	✓
Voice + Text Interaction	×	**Partial**	×	✓	×	×	**Partial**	✓
Speech-to-Speech Capability	×	×	×	×	×	×	×	✓
Order Taking through Conversation	×	×	×	×	×	×	**Partial**	✓
Order Tracking	×	×	×	×	×	×	✓	✓
Clothing & Fashion Recommendations	×	**Partial**	×	×	×	×	**Partial**	✓
Budget Optimization Suggestions	×	×	×	×	×	×	×	✓
Complaint Sentiment Analysis	×	×	×	×	×	×	×	✓
Admin Dashboard with Analytics	✓	✓	✓	✓	**Partial**	**Partial**	✓	✓

4. PROJECT OVERIEW/GOAL

The goal of the project is to develop an AI-powered Multilingual Customer Support and Order Management System for fashion e-commerce companies. Through an intelligent conversational interface, the proposed system will offer automated customer support in both Urdu and English, enabling customers to place orders, track order, receive clothing recommendations, and deal with complaints. Additionally, the system will have sentiment analysis for customer feedback, budget-aware fashion recommendations, and an intelligent admin dashboard for order management, complaint tracking, and customer behavior analysis.

The proposed system combines customer service, order tracking, recommendation systems, and sentiment analysis into a single platform, comparison to many current solutions provided by businesses like Shopify or Salesforce, which frequently focus on individual features. Using modern artificial intelligence and online technologies, the project's ultimate product will feature a web and app based, an AI chatbot interface, an order management database, and an admin analytics dashboard.

5. PROJECT DEVELOPMENT METHODLOGY / ARCHITECTURE

List of Libraries and Applications that will be used

- Python
- Jupyter Notebook
- VS Code / PyCharm
- React JS
- Node.js / Express.js
- MongoDB / MySQL
- TensorFlow / PyTorch
- NLP Libraries (NLTK / spaCy)
- Google Translate API (for multilingual support)
- Chart.js (for dashboard analytics)
- GitHub (for version control)
- Flutter
- Dart
- Android Studio

Module 1

Data Collection and Preparation

In this module, datasets related to **customer queries, fashion products, and clothing categories** will be collected and prepared. Data preprocessing techniques such as text cleaning, tokenization, and normalization will be applied to prepare the data for training AI models.

Module 2

AI Chatbot Development

This module focuses on building the **AI-powered multilingual chatbot** using Natural Language Processing techniques. The chatbot will be able to understand customer queries in **Urdu and English** and provide automated responses related to product information, order placement, and customer support.

Module 3

Fashion Recommendation System

In this module, a recommendation engine will be developed to suggest clothing items based on **user preferences, previous purchases, and budget constraints**. The recommendation system will help customers find suitable fashion products quickly.

Module 4

Order Management and Tracking System

This module will implement the functionality for **placing orders, updating order status, and tracking deliveries**. Customers will be able to check the status of their orders through the chatbot or the web interface.

Module 5

Complaint Handling and Sentiment Analysis

Customer complaints and feedback will be analyzed using **sentiment analysis techniques** to determine whether the feedback is positive, negative, or neutral. This will help businesses identify service issues and improve customer satisfaction.

Module 6

Admin Dashboard Development

This module will focus on developing a **smart admin dashboard** that allows administrators to monitor customer queries, track orders, analyze complaints, and view business insights through charts and reports.

Module 7

Frontend Development

The user interface of the web application will be developed using **React JS**. The interface will include chatbot interaction screens, order tracking pages, and recommendation displays for customers.

Module 8

Backend Development and Integration

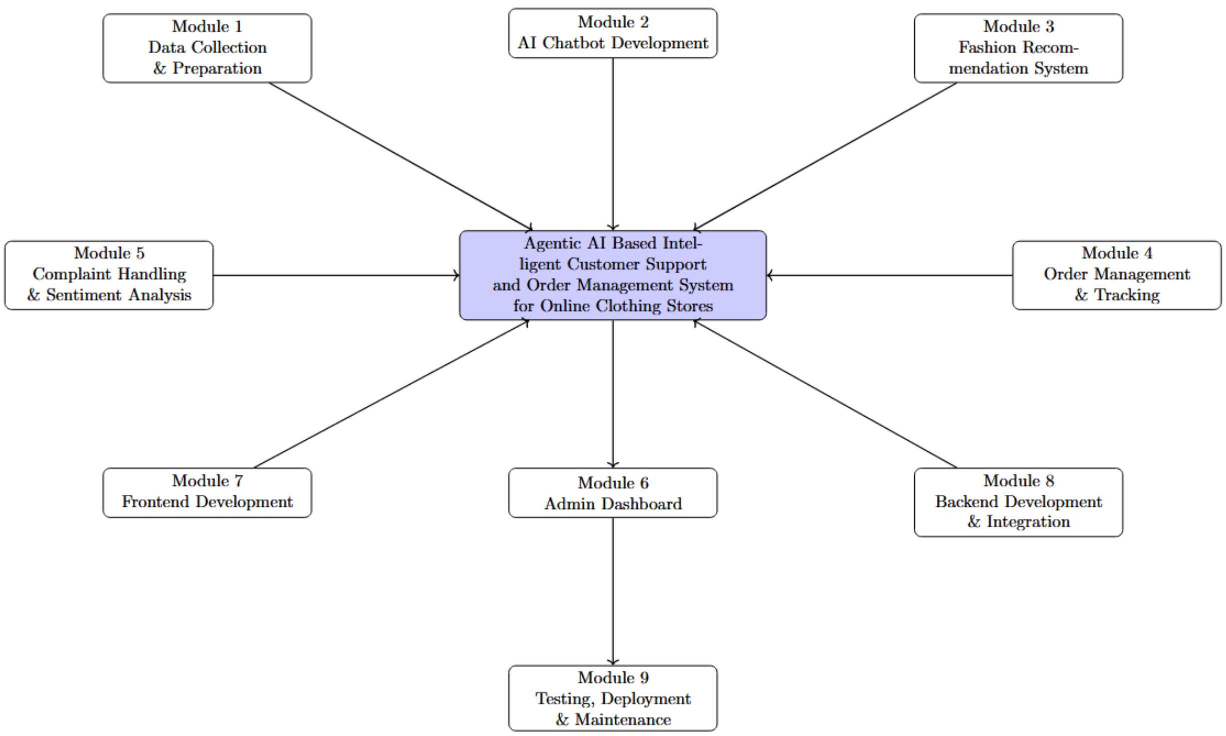
The backend system will be developed using **Node.js or Python frameworks**. This module will connect the chatbot, recommendation engine, database, and dashboard to ensure smooth data processing and communication between components.

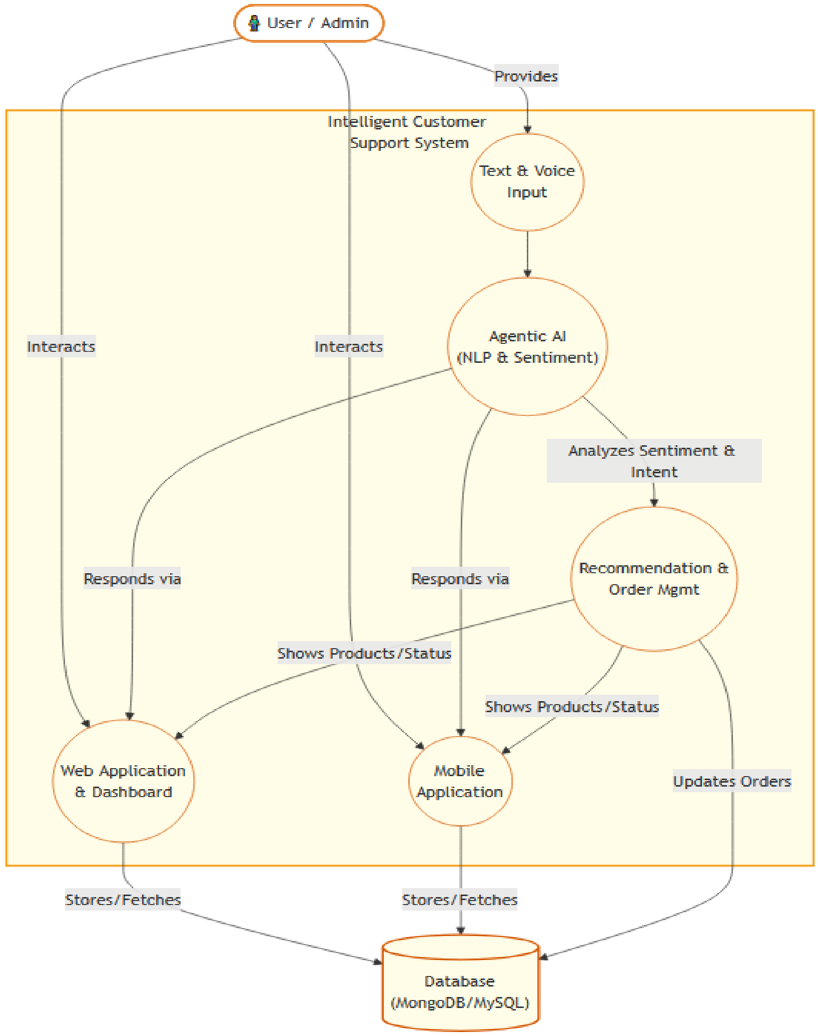
Module 9

Testing, Deployment, and Maintenance

In this final module, the entire system will be tested for functionality, usability, and performance. After debugging and optimization, the application will be deployed on a cloud platform. Future updates and bug fixes will be performed to maintain system performance.

Diagram for Modules



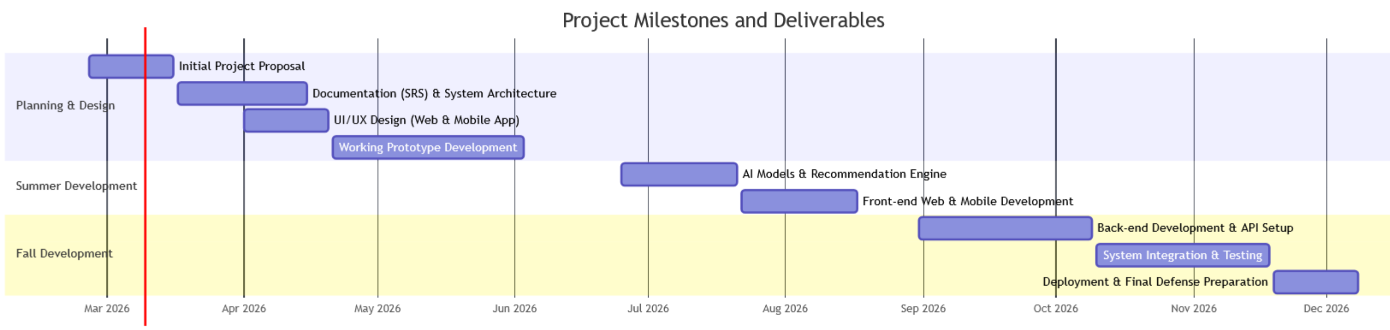


6. PROJECT MILESTONES AND DELIVERABLES

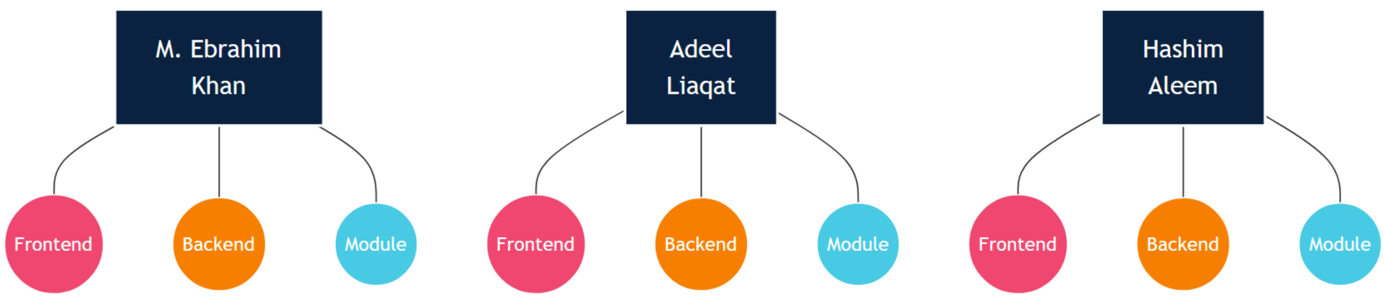
Task	Duration/Days	Start/Date	Finish/Date
Initial Project Proposal	20	25/02/2026	16/03/2026
Documentation (SRS) & System Architecture	30	17/03/2026	15/04/2026
UI/UX Design (Web & Mobile App)	20	01/04/2026	20/04/2026
Working Prototype Development	44	21/04/2026	03/06/2026
AI Models & Recommendation Engine	27	25/06/2026	21/07/2026
Front-end Web & Mobile Development	27	22/07/2026	17/08/2026
Back-end Development & API Setup	40	31/08/2026	09/10/2026

Task	Duration/Days	Start/Date	Finish/Date
System Integration & Testing	40	10/10/2026	18/11/2026
Deployment & Final Defense Preparation	20	19/11/2026	08/12/2026

Gantt Chart



7. WORK DIVISION



8. COSTING

1. Prototype Development:

- Software & AI: Open-source tools and locally hosted models.
- Infrastructure: Free databases and university/personal hardware.
- Total Cost: 0 PKR

2. Commercial Deployment:

- Hosting & Database: Domain, cloud servers, and secure cloud data storage – PKR 80,000
- AI API Services: Tokens for Agentic AI capabilities and language translation – PKR 50,000
- Security & Buffer: SSL certificates, firewall (WAF), and contingency funds – PKR 20,000
- Total Annual Cost: ~PKR 150,000

9. REFRENCES

[1] P. K. Singh and A. Sharma, "Conversational AI For E-Commerce Customer Service," *International Journal of Creative Research Thoughts (IJCRT)*, Feb. 2025.

[2] A. Hussain and O. Lemon, "Transforming Customer Engagement in Digital Commerce: The Role of Conversational AI Frameworks," *Journal of Information Systems Engineering and Management*, Nov. 2024.

[3] M. Awais and R. M. A. Nawab, "Towards a More Natural Urdu: A Comprehensive Approach to Text-to-Speech and Voice Cloning," *MDPI*, Oct. 2025.

[4] F. Ahmed and S. Ali, "Multilingual Transliteration based Urdu Chatbot using Rasa Framework," in *ResearchGate Conference Paper*, 2024.

[5] K. Reddy, V. Kumar, and T. Singh, "Smart Fashion Recommendation System Using CNN Res-Net 50," *International Journal of Novel Research and Development (IJNRD)*, vol. 8, no. 4, 2023.

[6] S. Gupta and R. Malhotra, "Fashion Recommendation System for E-Commerce using Deep Learning Algorithms," in *IEEE Conference Publication*, Jun. 2024.

[7] Y. Wang and J. Li, "Sentiment Analysis of E-Commerce Product Reviews Based on BiLSTM Model," in *IEEE Conference Publication*, Apr. 2024.

[8] L. Chen, H. Wu, and X. Zhang, "Sentiment Analysis for E-Commerce Product Reviews Using CNN-LSTM," in *IEEE Conference Publication*, Nov. 2024.

Apps References

https://www.zendesk.com/

https://www.intercom.com/

https://www.livechat.com/

https://omilia.com/

https://www.yalo.ai/

https://www.wati.io/